Practical 1.3: DevOps Essentials — Git, Docker, Docker Compose

Version control workflow, container management, multi-service deployment

Day 1 · 3 hours · Resilience Academy SDI Training

Reference: `crd.resilienceacademy.ac.tz` | Data: Tanga & Mwanza, Tanzania

Learning Objectives:

- Use core Git commands: clone, add, commit, push, pull, log
- Run and manage Docker containers from the command line
- Write and use a docker-compose.yml file
- Start a multi-container application with one command

Git Workflow

```
# Configure identity (if not already done)
git config --global user.name "Your Name"
git config --global user.email "you@example.com"

# Clone the GeoNode project repository
git clone https://github.com/GeoNode/geonode-project.git
cd geonode-project

# Inspect history
git log --oneline -10
git log --graph --oneline --all | head -20

# Check current branch
git branch -a

# Check status
git status

# Make a change and commit
echo "# My CRD Training Notes" > TRAINING.md
git add TRAINING.md
git status
git commit -m "Add training notes file"
git log --oneline -3
```

Git Branching

```
# Create and switch to a new branch
git checkout -b my-crd-customisation

# Make changes
echo "SITE_NAME=Tanzania CRD Training" >> TRAINING.md
git add TRAINING.md
git commit -m "Add site name setting"

# Switch back to main
git checkout main
```

```
# See branches
git branch
```

Docker Commands

```
# Pull images
docker pull ubuntu:22.04
docker pull nginx:latest
docker pull postgres:14

# List local images
docker images

# Run containers
# Interactive Ubuntu shell
docker run -it --rm ubuntu:22.04 bash

# Nginx web server (detached mode)
docker run -d --name my-nginx -p 8080:80 nginx
curl http://localhost:8080 | head -5

# View container logs
docker logs my-nginx
docker logs -f my-nginx # follow live

# List containers
docker ps # running
docker ps -a # all (including stopped)

# Stop and remove
docker stop my-nginx
docker rm my-nginx

# Inspect a container
docker inspect my-nginx
```

Docker Compose

```
# Create a project directory
mkdir ~/compose-exercise && cd ~/compose-exercise

# Create docker-compose.yml
cat > docker-compose.yml << 'EOF'
version: "3.8"
services:
  web:
    image: nginx:latest
    ports:
      - "8080:80"
    depends_on:
      - db
    volumes:
      - ./html:/usr/share/nginx/html

  db:
    image: postgres:14
```

```
    environment:
      POSTGRES_DB: training_db
      POSTGRES_USER: trainee
      POSTGRES_PASSWORD: resilience2024
    volumes:
      - pgdata:/var/lib/postgresql/data

volumes:
  pgdata:
EOF

# Create simple web page
mkdir html
echo "<h1>CRD Training - Docker Compose Test</h1>" > html/index.html

# Start all services
docker compose up -d

# Check status
docker compose ps

# View logs
docker compose logs

# Test the web server
curl http://localhost:8080

# Connect to PostgreSQL
docker compose exec db psql -U trainee -d training_db -c "\l"

# Stop all services
docker compose down

# Stop and remove volumes
docker compose down -v
```

Checkpoint:

- `git log` shows commits in the GeoNode repo
- Your TRAINING.md commit appears in the log
- `docker run hello-world` succeeds
- `docker compose ps` shows both web and db as "running"
- `curl http://localhost:8080` returns your HTML page